

Approval-based Participatory Budgeting

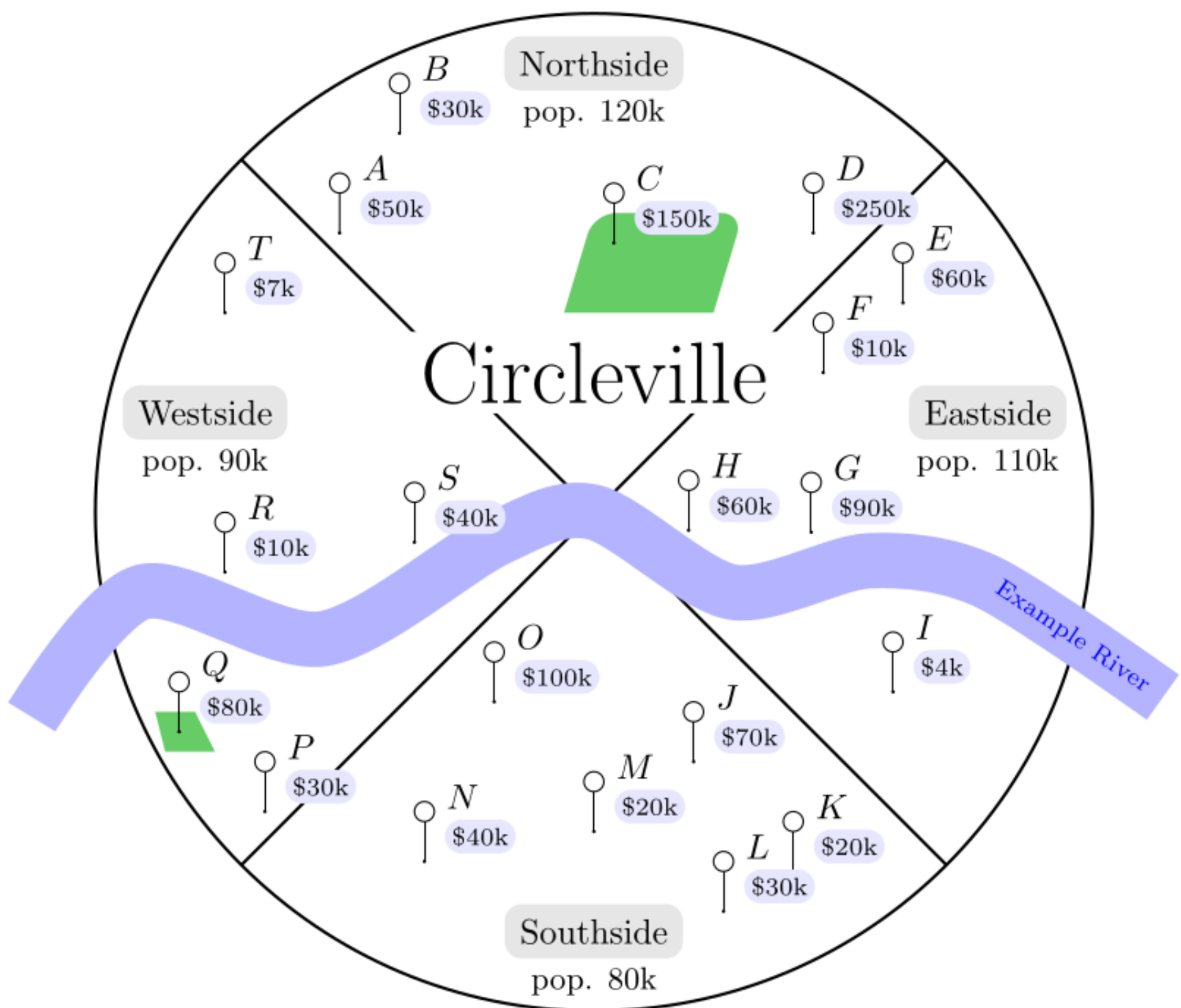


Figure 1 – The famous city of Circleville in Peters, Pierczyński, and Skowron (2021)

Method of Equal Shares

- Defined with a satisfaction function
- Allocate the budget to voters evenly
- Voters contribute to get their projects selected
- Iteratively selects project with smallest affordability

Example: budget of 8, 4 voters and 3 projects:

	1	2	3	4	cost	decision
p_1	✓	✓	✓	✗	3	✓
p_2	✗	✗	✗	✓	3	✓
p_3	✓	✗	✗	✓	4	✗

Some nice axiomatic guarantees:

- EJR-1+: a P -cohesive group of voters gets $\text{sat}(P)$ up to one project
- Priceability: selected projects can be afforded by supporters

MES performs well on real world data

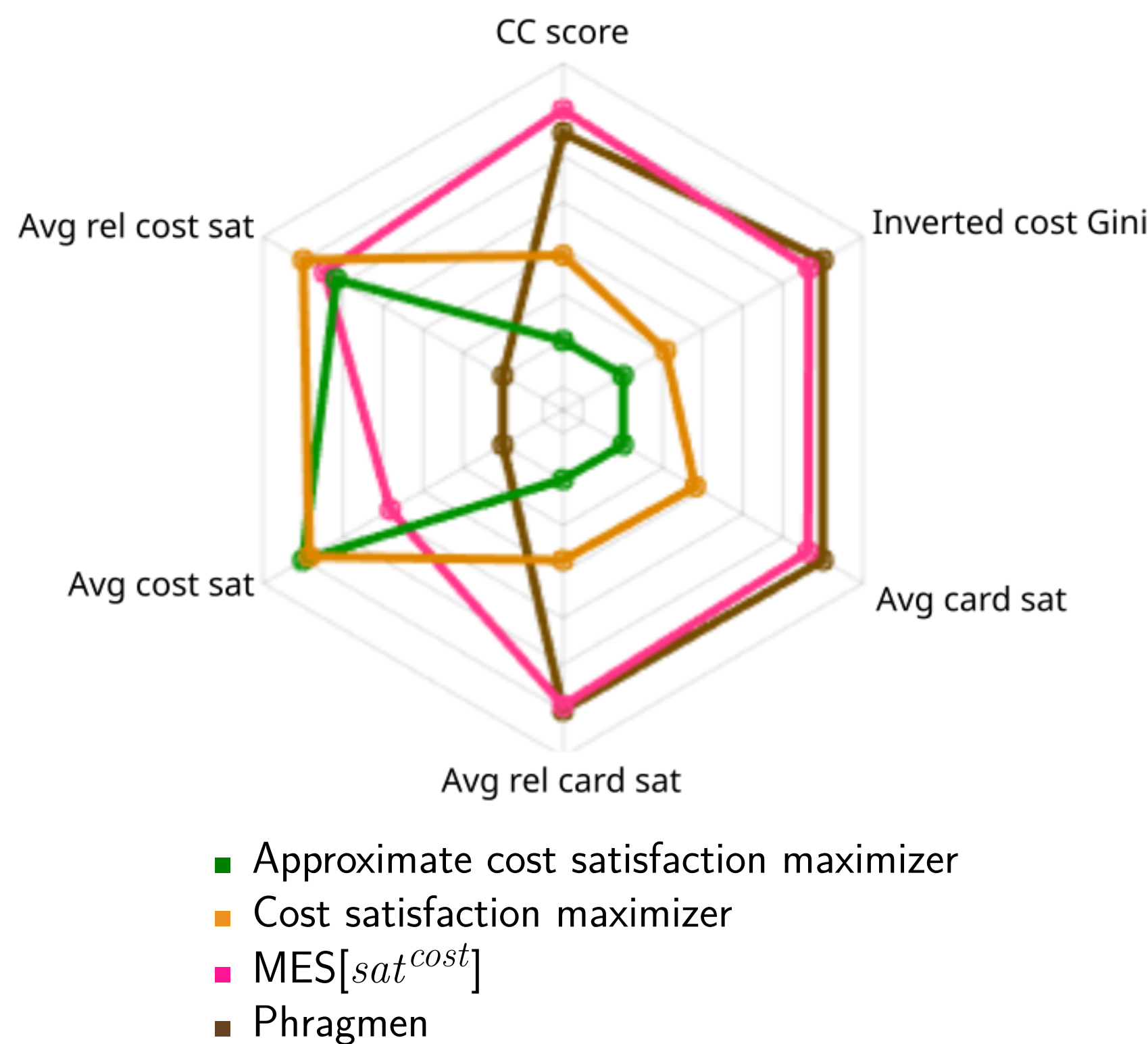


Figure 2 – Comparison of different PB rules on <https://pabuviz.org>

Fair Share

Introduced by Maly et al. (2023), inspired from fair allocation theory.

$$\text{share}_i(P) = \sum_{p \in P} \frac{c(p)}{|j \in \mathcal{N}, p \in A_j|}$$

Then, the fair share of a voter $i \in \mathcal{N}$ is

$$\text{fairshare}(i) = \min(b/n, \text{share}_i(A_i))$$

Fair Share is not always possible to satisfy

- Weaker versions: local FS and EJS
- Distance to FS (L1 distance)
- FS capped ratio: portion of fair share

MES[share]: satisfies FS and EJS

Did not outperform MES[cost] on pabulib data

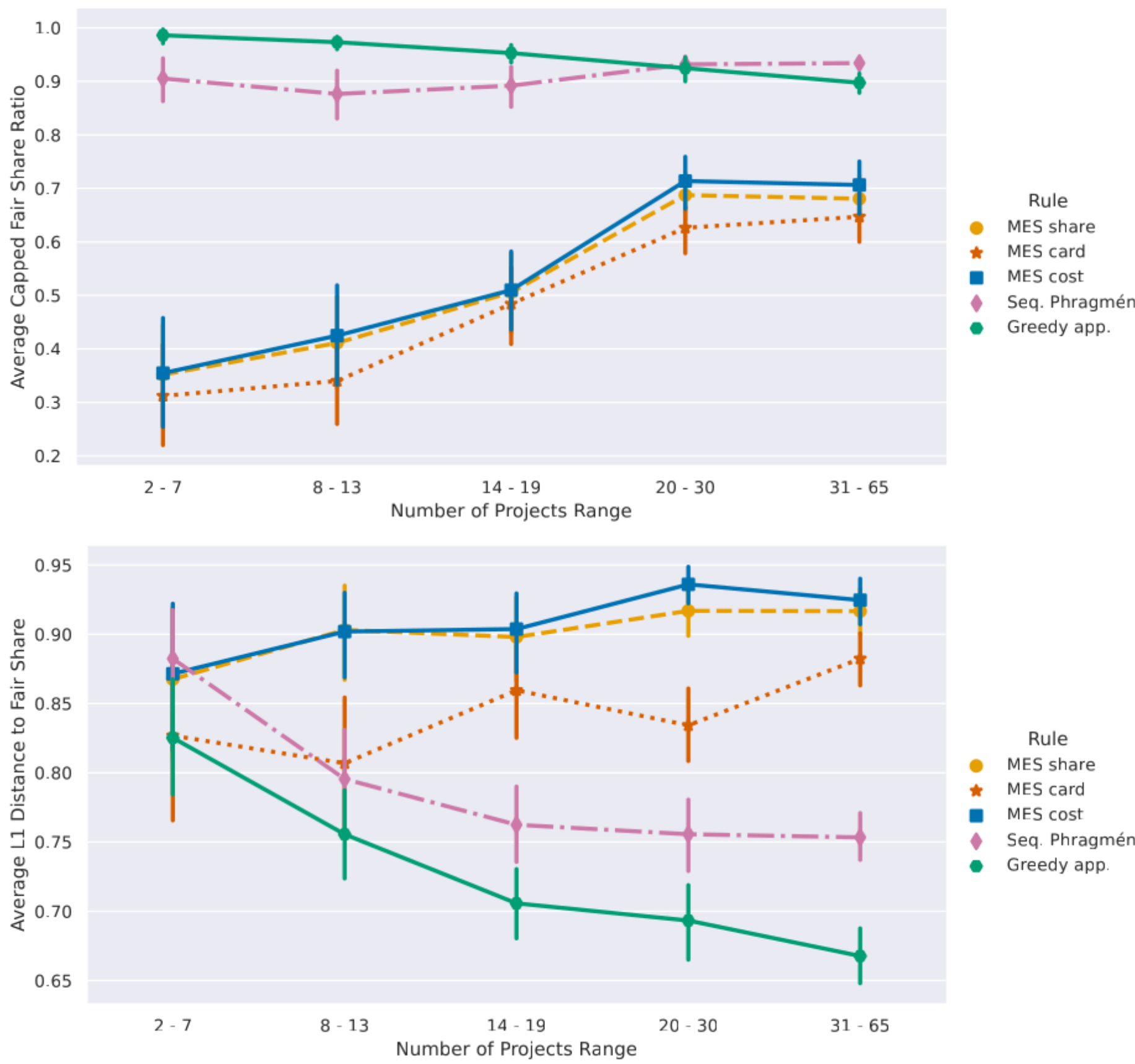


Figure 3 – Average capped FS ratio and distance to FS from Maly et al. (2023)

Why MES[share] does not outperform MES[cost]

p_3, p_4, p_5 appear 10 times, budget of 8.

	1	2	3	4	cost	MES[cost]	MES[share]
p_1	✓	✓	✓	✗	3	1	1
p_2	✗	✗	✓	✓	3	1	0
p_3	✓	✗	✗	✗	0.1	10	10
p_4	✗	✓	✗	✗	0.1	10	10
p_5	✗	✗	✗	✓	0.1	0	10

Problem with MES[cost]: 4 has to pay too much for p_2 .

Problem with MES[share]: we select projects with low satisfaction.

Solution: spread the 'debt' of voter 3.

DMES: computes the affordability of a set of projects and selects p s.t. $\pi \cup \{p\}$ has min affordability.

Inspired by Sánchez-Fernández et al. (2022)

	1	2	3	4	cost	DMES[cost]
p_1	✓	✓	✓	✗	3	1
p_2	✗	✗	✓	✓	3	1
p_3	✓	✗	✗	✗	0.1	7
p_4	✗	✓	✗	✗	0.1	7
p_5	✗	✗	✗	✓	0.1	6

DMES[cost] and DMES[share]

About DMES[cost]:

- Keeps axiomatic guarantees (EJR, priceability)
- Selects larger sets of projects
- Can provide unfair allocations
- Harder to compute than MES (still poly time)

Budget of 8.

	1	2	3	4	cost	DMES[cost]	DMES[share]
p_1	✓	✓	✓	✗	5	✓	✓
p_2	✗	✗	✓	✓	3	✓	✗
p_3	✓	✗	✗	✗	1/3	✗	✓
p_4	✗	✓	✗	✗	1/3	✗	✓
p_5	✗	✗	✗	✓	2	✗	✓

Ongoing work

Find an exact algorithm for computing dynamic affordability

LP programming for now:

$$\begin{aligned} \min & q \\ \text{s.t.} & \\ & \forall i \in \mathcal{N}, p' \in W \cup \{p\}, \gamma_i(p')/\text{sat}(p') \leq q \\ & \forall i \in \mathcal{N}, p' \notin A_i^+ \cup (W \cap \{p\}), \gamma_i(p') = 0 \\ & \forall i \in \mathcal{N}, \sum_{p' \in (W \cup \{p\}) \cap A_i} \gamma_i(p') \leq b/n \\ & \forall p' \in W \cup \{p\}, \sum_{i \in \mathcal{N}(p')} \gamma_i(p) = c(p') \\ & \alpha \leq 1 \end{aligned}$$

Idea: Network flows like Sánchez-Fernández (2023).

Define resolute version of DMES based on BOS MES from Papasotiropoulos et al. (2024):

$$\begin{aligned} \min & q/\alpha \\ \text{s.t.} & \\ & \forall i \in \mathcal{N}, p' \in W \cup \{p\}, \gamma_i(p')/\text{sat}(p') \leq q \\ & \forall i \in \mathcal{N}, p' \notin A_i^+ \cup (W \cap \{p\}), \gamma_i(p') = 0 \\ & \forall i \in \mathcal{N}, \alpha \times \sum_{p' \in (W \cup \{p\}) \cap A_i} \gamma_i(p') \leq b/n \\ & \forall p' \in W \cup \{p\}, \sum_{i \in \mathcal{N}(p')} \gamma_i(p) = c(p') \\ & \alpha \leq 1 \end{aligned}$$

Experiments, measuring voter's satisfaction and fairness

Runned MES[cost], DMES[cost] and DMES[share] without completion.

Rules comparison with normalized data (Gini and FS dist are inverted)

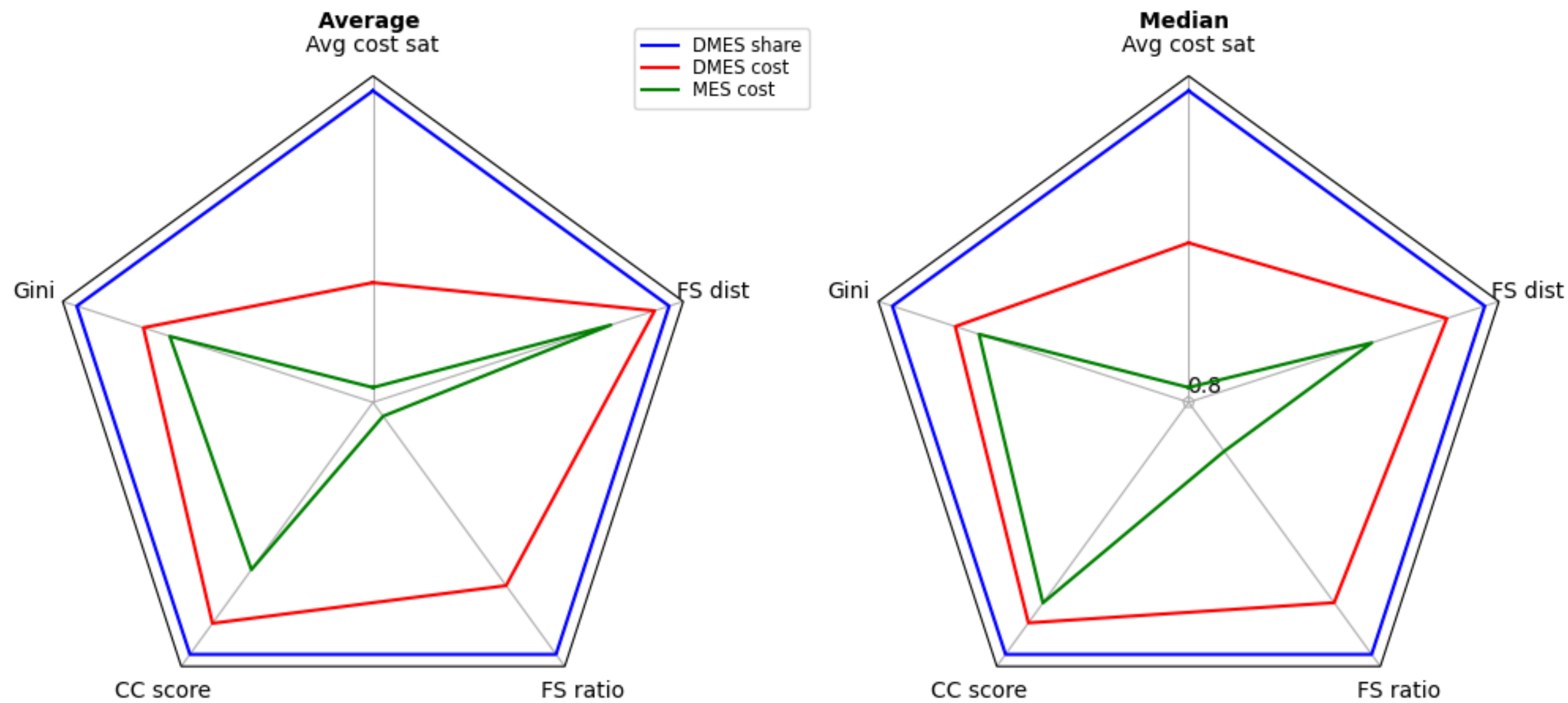


Figure 4 – Comparison of MES variations without completion

Rules comparison with normalized data (Gini and FS dist are inverted)

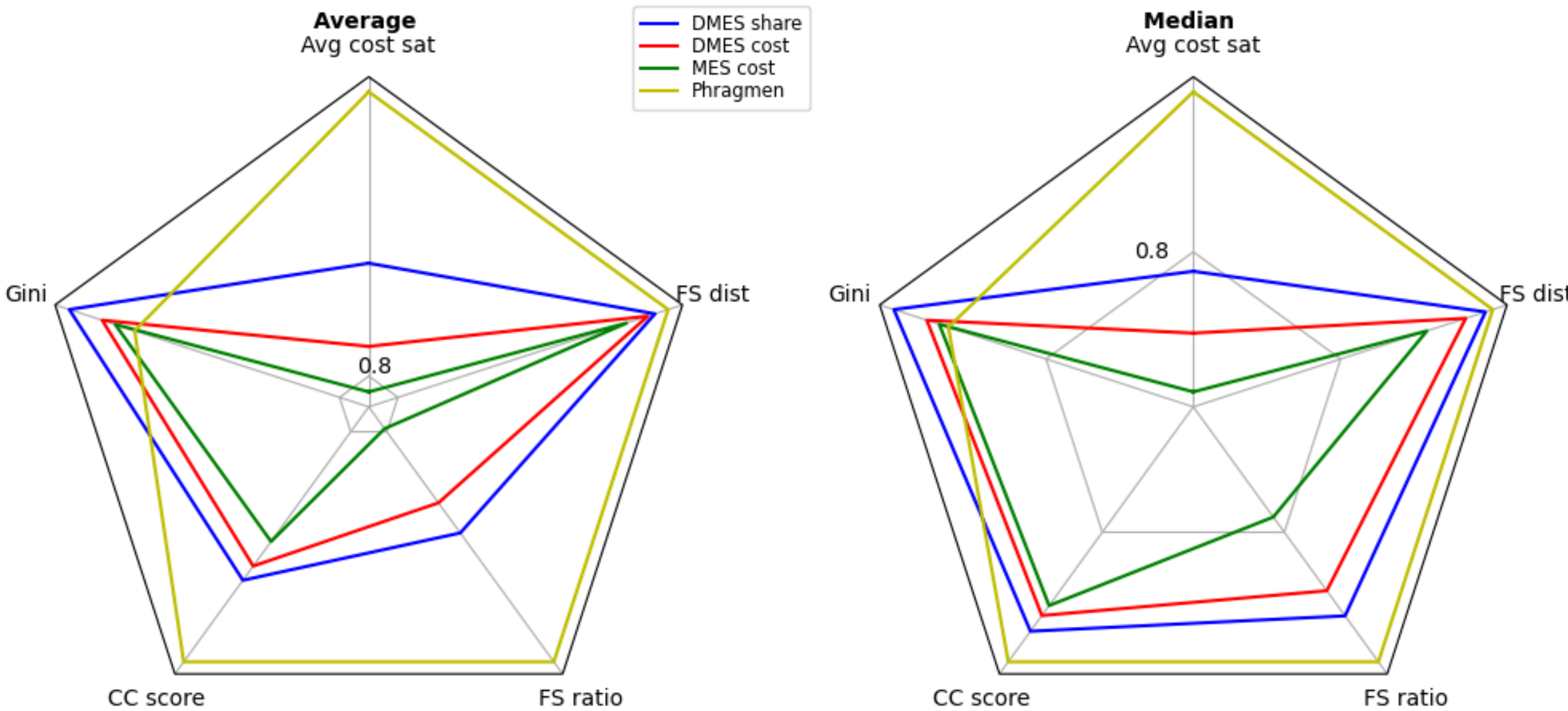


Figure 5 – Comparison of MES variations with Phragmen

Acknowledgments

Thanks to Dominik Peters and Piotr Skowron for useful discussions and comments.

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